Indian Institute of Technology Kanpur
CS771: Introduction to Machine Learning

Assignment 1 Report

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1 Problem 1. Mathematical Derivation for Prediction

To express the Hamming PUF operations as a linear model, we map the binary challenge bits $c \in \{0,1\}^{32}$ and secret bits $s \in \{0,1\}^{32}$ into the **bipolar domain** $\{-1,1\}$. *(where $0 \mapsto 1$ and $1 \mapsto -1$)*

Let

$$x_i = 1 - 2c_i$$

and

$$y_i = 1 - 2s_i.$$

In this domain, the XOR operation $m_i = c_i \oplus s_i$ can be expressed as algebraic multiplication:

$$1 - 2m_i = x_i y_i$$

which implies

$$m_i = \frac{1 - x_i y_i}{2}.$$

Binary Domain			Bipolar Domain		
c_i	s_i	$c_i \oplus s_i$	$x_i = 1 - 2c_i$	$y_i = 1 - 2s_i$	$m_i = \frac{1-x_i y_i}{2}$
0	0	0	1	1	0
0	1	1	1	-1	1
1	0	1	-1	1	1
1	1	0	-1	-1	0

We can now substitute this into the equations for the partial Hamming weights. For the even locations:

$$h_e = \sum_{i=0}^{15} m_{2i} = \sum_{i=0}^{15} \frac{1 - x_{2i} y_{2i}}{2} = 8 - \frac{1}{2} \sum_{i=0}^{15} x_{2i} y_{2i}.$$

For the odd locations:

$$h_o = \sum_{j=0}^{15} m_{2j+1} = \sum_{j=0}^{15} \frac{1 - x_{2j+1} y_{2j+1}}{2} = 8 - \frac{1}{2} \sum_{j=0}^{15} x_{2j+1} y_{2j+1}.$$

The PUF response is 1 if $h_e h_o \geq t$. Expanding the product $h_e h_o$:

$$h_e h_o = \left(8 - \frac{1}{2}\sum_{i=0}^{15} x_{2i} y_{2i}\right)\left(8 - \frac{1}{2}\sum_{j=0}^{15} x_{2j+1} y_{2j+1}\right).$$

Thus,

$$h_e h_o = 64 - 4\sum_{i=0}^{15} x_{2i} y_{2i} - 4\sum_{j=0}^{15} x_{2j+1} y_{2j+1} + \frac{1}{4}\sum_{i=0}^{15}\sum_{j=0}^{15} (x_{2i} x_{2j+1})(y_{2i} y_{2j+1}).$$

The inequality condition for the response being 1 is therefore

$$\begin{aligned} 64 - 4\sum_{i=0}^{15} x_{2i} y_{2i} - 4\sum_{j=0}^{15} x_{2j+1} y_{2j+1} + \frac{1}{4}\sum_{i=0}^{15}\sum_{j=0}^{15} (x_{2i} x_{2j+1})(y_{2i} y_{2j+1}) - t \geq 0 \\ \implies 64 - 4x_E^\top y_E - 4x_O^\top y_O + \frac{1}{4}(x_E^\top y_E)(x_O^\top y_O) - t \geq 0 \end{aligned}$$

where

$$x_E = (x_0, x_2, \ldots, x_{30})^\top, \qquad x_O = (x_1, x_3, \ldots, x_{31})^\top,$$

and similarly for y_E and y_O.

Defining

$$\phi(c) = \begin{bmatrix} x_E \\ x_O \\ x_E \otimes x_O \end{bmatrix}, \qquad w = \begin{bmatrix} -4y_E \\ -4y_O \\ \frac{1}{4}(y_E \otimes y_O) \end{bmatrix}, \qquad b = 64 - t,$$

the inequality can be written compactly as

$$w^\top \phi(c) + b \geq 0.$$

This is a linear inequality of the form $\boldsymbol{w}^\top \phi(c) + b \geq 0$. We can separate the terms that depend purely on the challenge c (the features $\phi(c)$) from the terms that depend on the secret s and threshold t (the weights $\boldsymbol{w}$ and bias b).

Given a challenge c, first compute $x_k = 1 - 2c_k$. The feature vector $\phi(c)$ consists of the following components:

- x_{2i} for $i \in \{0, \ldots, 15\}$ (the even bits)
- x_{2j+1} for $j \in \{0, \ldots, 15\}$ (the odd bits)
- $x_{2i} x_{2j+1}$ for $i, j \in \{0, \ldots, 15\}$ (all pairwise products of even and odd bits)

The corresponding weight vector $\boldsymbol{w}$ consists of:

- $-4y_{2i}$ corresponding to the even-bit features,
- $-4y_{2j+1}$ corresponding to the odd-bit features,
- $\frac{1}{4} y_{2i} y_{2j+1}$ corresponding to the pairwise-product features.

The bias term is $b = 64 - t$. Because the inequality exactly matches $\boldsymbol{w}^\top \phi(c) + b \geq 0$, the response can be perfectly modeled as:

$$r(c) = \frac{1 + \text{sign}(\boldsymbol{w}^\top \phi(c) + b)}{2}.$$

2 Problem 2. Dimensionality Calculations

The dimensionality D of the linear model is determined by the number of unique elements in the derived feature map $\phi(c)$. Breaking down the elements of $\phi(c)$:

1. **Even terms:** There are 16 even-indexed bits $(x_0, x_2, \ldots, x_{30})$, contributing **16 dimensions**.
2. **Odd terms:** There are 16 odd-indexed bits $(x_1, x_3, \ldots, x_{31})$, contributing **16 dimensions**.
3. **Cross-product terms:** We take the product of every even bit with every odd bit, $x_{2i}x_{2j+1}$. Since there are 16 even bits and 16 odd bits, there are $16 \times 16 = 256$ unique cross terms, contributing **256 dimensions**.

Summing these contributions,

$$D = 16 + 16 + 256 = 288.$$

Therefore, a linear model requires exactly **288** dimensions to perfectly predict the responses for a 32-bit Hamming PUF.

3 Problem 4. Experimental outcomes and hyperparameter analysis

We evaluated LinearSVC and LogisticRegression to break the Hamming PUF using the 288-dimensional embedding. Results are reported on the public dataset.

3.1 Impact of loss function (LinearSVC)

We compared Hinge and Squared Hinge loss functions. As shown in Table 1, Squared Hinge loss function is more efficient for this dataset while maintaining high accuracy.

The results indicate that both loss functions achieve nearly identical training and test accuracies (around 99%), suggesting that the transformed feature space makes the Hamming PUF problem almost linearly separable. However, Squared Hinge loss function significantly reduces the training time from 1.224 s to 0.287 s. This is because the Squared Hinge loss function is smoother, differentiable, and easier to optimize, leading to much faster convergence while maintaining similar accuracy.

Overall, Squared Hinge loss function is preferable for this question because it provides the same accuracy as Hinge loss function with significantly lower training time.

Table 1: Comparison of loss functions in LinearSVC

Loss	Train Acc (%)	Test Acc (%)	Time (s)
Hinge	99.88	99.88	1.224
Squared Hinge	99.92	99.88	0.287

3.2 Impact of regularization strength (C)

As shown in Table 2, increasing C from 0.001 to 0.1 significantly improves both training and test accuracy for LinearSVC and Logistic Regression. At very small values of C, strong regularization causes underfitting, resulting in lower accuracy.

For LinearSVC, the test accuracy increases from 96.80% to approximately 99.88% as C increases. Logistic Regression exhibits a similar trend, with test accuracy improving from 93.56% to over 98%. Increasing C beyond 0.5 doesn't provides any noticeable improvement in accuracy, indicating that the model has already captured the underlying decision boundary.

These results suggest that medium values of C offer the best balance between regularization and accuracy.

Table 2: Effect of Regularization Strength (C) on Accuracy and Training Time

	LinearSVC			LogisticRegression		
C	Train Acc	Test Acc	Time (s)	Train Acc	Test Acc	Time (s)
0.001	98.55%	96.80%	0.186	95.73%	93.56%	0.031
0.1	99.93%	99.80%	0.267	99.23%	98.16%	0.074
0.5	99.92%	99.88%	0.268	99.25%	98.16%	0.072
1.0	99.95%	99.88%	0.259	99.25%	98.16%	0.077
10.0	99.97%	99.88%	0.262	99.25%	98.12%	0.063

3.3 Impact of tolerance (tol)

Tolerance (tol) acts as a stopping criterion for the optimization algorithm.

The results in Table 3 show that decreasing the tolerance improves accuracy at the cost of increased training time. A larger value tolerance causes the optimization process to terminate earlier, which resulted in lower accuracy.

This effect is more noticeable for Logistic Regression, where a tolerance of 0.1 leads to only 48.00% test accuracy. Reducing the tolerance to 0.01 increases the test accuracy to 98.16%, indicating that additional optimization iterations are necessary to obtain a good accuracy.

Further decreasing the tolerance to 0.0001 produces only marginal improvements while increasing training time. Therefore, a tolerance value around 0.01 appears to provide a good trade-off between accuracy and training time.

Table 3: Effect of Tolerance (tol) on Accuracy and Training Time

	LinearSVC			LogisticRegression		
`tol`	Train Acc	Test Acc	Time (s)	Train Acc	Test Acc	Time (s)
0.1	99.17%	98.16%	0.215	47.16%	48.00%	0.013
0.01	99.92%	99.88%	0.262	99.25%	98.16%	0.045
0.001	99.99%	99.84%	0.290	99.91%	99.52%	0.089
0.0001	99.99%	99.84%	0.300	99.96%	99.56%	0.130
1e-05	99.99%	99.84%	0.327	99.96%	99.52%	0.165

3.4 Impact of penalty ($L1$ vs $L2$)

As shown in Table 4, both $L1$ and $L2$ regularization achieve nearly identical test accuracies. However, the training time differs significantly between the two penalty types.

For LinearSVC, training time decreases from $13.705\,\mathrm{s}$ with $L1$ to only $0.261\,\mathrm{s}$ with $L2$. A similar trend is observed for Logistic Regression, where $L2$ trains considerably faster while maintaining the same accuracy.

Therefore, $L2$ regularization is preferred because it achieves similar accuracy while requiring significantly less training time.

Table 4: Effect of Penalty Type on Accuracy and Training Time

	LinearSVC			LogisticRegression		
Penalty	Train Acc	Test Acc	Time (s)	Train Acc	Test Acc	Time (s)
L1	100.00%	99.88%	13.705	100.00%	99.64%	2.521
L2	99.92%	99.88%	0.261	99.99%	99.64%	0.162

4 Problem 5. Impact of training set size on accuracy

To determine the minimum number of Challenge-Response Pairs (CRPs) required to break the Hamming PUF effectively, we conducted experiments with varying training set sizes ranging from 100 to 7500 points. For each size, we performed 10 independent trials, selecting training points randomly without replacement from the original set while keeping the test set fixed.

4.1 Performance across dataset sizes

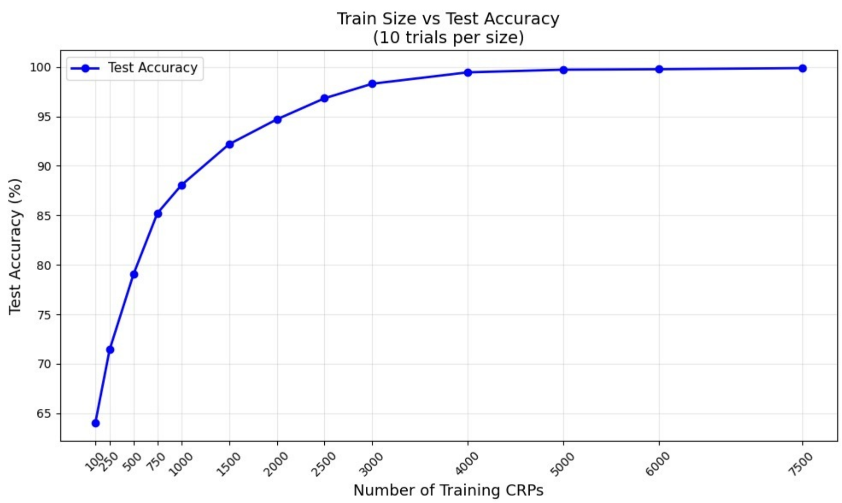

Figure 1: training set size vs test accuracy over 10 trials per size.

For each training size, we randomly sampled training points from the original training set and repeated the experiment 10 times. The reported accuracy is the average test accuracy across these 10 trials.

As shown in the above curve, accuracy increases rapidly when training size is small, but the improvement per additional CRP decreases as the training set grows larger.

The experimental results indicate a logarithmic-like growth in accuracy relative to the number of training points. While the full dataset contains 7500 points, near-perfect accuracy is achieved much earlier.

4.2 Minimum crps for accuracy thresholds

Based on our empirical analysis, the following minimum number of CRPs are required to reach specific accuracy milestones:

- **95% Accuracy:** Approximately **2100 CRPs** (achieved 95.30% in trials).
- **97% Accuracy:** Approximately **2600 CRPs** (achieved 97.08% in trials).
- **99% Accuracy:** Approximately **3500 CRPs** (achieved 99.18% in trials).

Beyond 3500 points, the accuracy gains become marginal, with the model saturating near 99.8% at the full 7500 points. These thresholds suggests that the proposed 288-dimensional feature map is highly sample-efficient. Less than half of the available 7500 CRPs are required to achieve over 99% test accuracy.